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MALNUTRITION AND REFORM IN E-GOVERNANCE

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Abstract --- *The Personalized Public Distribution System (PDS) presented in this paper revolutionizes food subsidy allocation by integrating detailed health metrics such as BMI, MUAC, blood glucose, serum protein, and micronutrient levels (e.g., vitamin A, zinc, iodine). Unlike traditional systems that allocate food based on broad demographic categories like income, this system tailors food subsidies to individual health needs, addressing both macronutrient and micronutrient deficiencies. This personalized approach ensures that low-income and nutritionally vulnerable populations receive targeted, nutrient-dense foods, contributing to the achievement of Sustainable Development Goal 2 (Zero Hunger) and SDG 10 (Reduced Inequality). The system's dynamic nature allows for real-time health tracking and adaptive food distribution, optimizing health outcomes while minimizing waste. Future enhancements, such as longitudinal health data integration and predictive analytics, can further refine personalization and improve the system's impact. This innovative approach has the potential to transform public health nutrition, reduce malnutrition, and accelerate global progress toward sustainable development goals.*

Keywords --- *Personalized Public Distribution System (PDS), Health Metrics, Nutrient-Dense Foods, Macronutrient Deficiencies, Micronutrient Deficiencies, Sustainable Development Goals (SDG), Zero Hunger, Reduced Inequality, Food Subsidy Allocation, Real-Time Health Tracking.*

I. INTRODUCTION

A. Background of malnutrition

India scores 28.7, based on a scale of 0–100 points with 0 point representing no hunger and 100 is the worst situation. For the year 2023, the GHI ranks India at position 111 out of 125 countries. This value puts India in the serious category of hunger with a score of 28.7. People who do not get enough food will suffer from hunger problems, and which over a long term can become malnutrition. Information to note here, a person can be malnourished due to some other factors and have nothing to do with hunger. People with the ability to consume enough foods may be poorly nourished since they do not take food that supply the amount needed nutrient, vitamins, minerals and other essentials for proper nourished growth and wellness. Malnutrition is a public health issue that affects millions worldwide and has different levels of impact from the area. As part of the GHI, both undernutrition and overnutrition are brought together into one index that measures hunger as well as malnutrition. The ranking that this GHI comes up with has countries ranked according to their level of hunger, worst to best. It utilizes parameters such as undernourishment and three children's parameters: stunting in children, wasting in children, and under-five mortality.

B. Regional Disparities in Malnutrition

Sub-Saharan Africa is the region with the highest levels of undernutrition and stunting. Most countries have high GHI measures, which indicates severe hunger and malnutrition. Some causes of this are long-standing poverty, restricted access to quality healthcare, and recurrent food shortages due to climatic factors and conflicts. Chronic food insecurity and acute emergencies that cause wasting are the major causes of stunting. Such conditions have caused heavy backlashes, especially child mortality rates and inadequate cognitive and physical development. The region under South Asia, comprising Bangladesh, India, and Nepal, has a significantly high rate of malnutrition. The GHI also shows that this region has the greatest prevalence of underweight children and the largest percent of stunted children. All of it contributes toward poor sanitation, a lack of resources for maternity and child health care, and pervasive poverty. To a certain degree, South Asian malnutrition supports the cycle of poverty and ill health that leads to poorer educational and economic outcomes.

C. Types of Malnutrition

1. Undernutrition

The common types include stunting, wasting, and being underweight. Stunting develops from chronic nutrition deficiencies at critical growth periods during childhood, leading to permanent reductions in height. Wasting, mostly associated with acute malnutrition, leads to great weight loss and is most found in emergency settings or crises. Being underweight reflects both acute and chronic undernutrition and thus represents an important indicator of overall nutritional deficiency.

2. Overnutrition

It is caused by calorie intake that is greater than the body's needs and a diet lacking in quality, which leads to a plethora of health issues such as heart problems, diabetes, and other forms of cancer. Increased overnutrition and undernutrition patterns are in some areas changing from undernutrition to overnutrition, which will display involvement of society in integral multi-faceted approaches that will balance both extremes of malnutrition.

D. Consequences of malnutrition

Childhood malnutrition eventually leads to both mental and physical catch-up growth, increased susceptibility to infections, and increased mortality. It impacts on school performance and ultimately productivity in the labor

market, thus forming an unending cycle of poverty with ill health. For adults, the impact persists through reductions in productivity, increased costs for health care, and generally lower quality of life. Its aftermath multiplies costs of chronic disorders and is also likely to have low productivity with financial instability.

E. Causes of Malnutrition

There are specific causes of malnutrition that are influential in character as they affect different regions and people differently. Most people agree that the major cause of malnutrition is the economic inequality among people. Low-income groups do not eat well because they do not have access to good quality food but rather the cheap and less nutritious foods which leads to under-nutrition or obesity problems. In fact, it has been estimated that in India, seventy-four percentage of the population is unable to afford a healthy diet. The economic pressure makes it impossible for most of the people to get all the necessary nutrients. As a result of the change in dietary habits, most of the people take a lot of processed along with sugary foods and very little of the balanced diet which is rich in nutrients. Due to the growing obesity epidemic, there has been a notable deficiency of key vitamins and minerals in the diets especially in the developing nations such as India, including Iron, Vitamin A, and Zinc. Still, the greater and increasing amounts of processed foods intake have also been linked with the even greater levels of prevalence of obesity among populations that were previously in the low nutrition rate.

Inadequate sanitation and hygiene practices contribute to the increase of malnutrition levels, especially among children. With poor sanitation, there are many infections that do not allow the nutrient absorption of the essential nutrients in the body. In India, this is a notable concern as only 69% of the households have access to hygiene and sanitation facilities which helps in controlling spread of diseases hence malnutrition. The availability of primary healthcare services is the other major causative factor. A number of these services include vaccination and antenatal care, and in this case however, their rural population does not avail of these services. This makes them more prone to malnourishment especially when coupled with lack of education on the healthy ways of eating. However, there is often an inverse relationship between health care access and utilization.

Government-led initiatives aimed at curbing malnutrition do experience sometimes slow implementation and inconsistent distribution. The lack of adequate operational mechanisms ensures that most of the local nutrition childcare services in India offer severely inadequate nutrition to the children. As a result, the problems in combating malnutrition are not only non-existent but even make attaining the goal of combating malnutrition counterproductive and therefore food and other resources cannot be utilized in the most appropriate manner.

F. Government Policies and Solutions to Address Malnutrition

The issue of malnutrition has been the subject of many policies and programs created by various governments worldwide. They have launched several interventions in the direction of ensuring that everyone, more so those in poor households, receives adequate and satisfactory nutrition. Such include food assistance programs, school lunches and other enrolment campaigns targeting young children. School meal programs prepare and serve infants and young children at least one nutritious meal per day, a very big chunk of their daily nutrition. However, in terms of quality and nourishment, there exists a lot of difference between one school feeding programme and another. For some, there are too many logistical problems, very limited resources or even corruption that prevents access to the services for the deserving population. The Public Distribution System, which is implemented by the government in many countries, is a scheme where essential commodities like rice, wheat and sugar are made available to the Below Poverty Line households at controlled prices. PDS does assist in bridging the bare minimum caloric consumption requirements but is very much limited in addressing the expansive deficit of nutrition requirements. The saccharine and fat laden processed food items made available are many in number but limited in proteins, vitamins and other essential nourishing elements. The other limitation is, there is systemic wastage, graft, and misapplication of the aid where the food intended for the needy does not get to them. To put it more clearly, even though these government meals have helped curb the cases of malnutrition inhibiting factors remain.

G. Traditional Public Distribution Systems (PDS)

1. Overview of Public Distribution Systems

Public Distribution Systems are government-run program. These Programs are implemented to guarantee food security and provide low-income and vulnerable groups affordable food. PDS's main goal is to provide basic food supplies at reasonable price, especially to people who can't buy them on market. PDS systems have changed throughout time in response to shifting governmental goals and socioeconomic circumstances. Food is sourced centrally and delivered via a network of regional and local centres, with system being decentralised to effectively manage local distribution. The main objectives of PDS are:

- I) Ensuring Food Security
- II) Supporting Low-Income Families
- III) Stabilizing Food Prices

2. Structure and Implementation of PDS

2.1. Centralized Procurement

Large amount of staple goods like rice, wheat and sugar are purchased by the government straight from the farmers and markets. Special entities like Food Corporation of India (FCI) oversee this purchasing process. To guarantee that farmers are paid fairly for their goods, the government defines a minimum price, sometimes referred as Minimum Support Price or MSP. This guarantees a consistent supply of food and aids in stabilising farmer's income.

2.2. Storage and Warehousing

Food is kept in government-owned warehouses once it is purchased. Because they preserve food until it is required, these storage places are essential. The purchased goods may be kept fresher and less damaged by storing it properly. Large amount of food can be kept safe in warehouse using their equipment.

2.3. Distribution Channels

Food is transported from central warehouse to local and regional distribution centres. Government vehicles or hired vehicles are used for this transportation. The main goal is to transport the food gets to local regions quickly and effectively. Once the food reaches the local distribution centres it is given to eligible consumers. FPS is usually managed by private retailers that have agreements with the government. They help families with low incomes afford the food by offering affordable rates.

2.4. Distribution Process

Fair Price Shops are little stores where customers may pick up the food goods that are allotted to them. The stores are tasked of providing food to customers based on their ration cards. Usually, the distribution is determined by the socioeconomic condition and size of the household.

3. Eligibility and Beneficiary Identification

Income levels and socioeconomic position are considered while defining the eligibility. Beneficiaries are determined via household surveys, income certificates, or ration cards in various systems. Based on the income they divided into Above Poverty Line (APL) and Below Poverty Line (BPL). Based on that the goods are distributed.

4. Monitoring and Evaluation

Regular inspections are carried out to make sure that the supplied food satisfies quality expectations. This includes checking for contamination and ensuring that the food is not expired. Periodic evaluations are taken to spot the problems like leakage, diversion and corruption. In many PDS system customer can file a complaint or contribute feedback on the food's quality.

5. Limitations of Traditional PDS

Staple foods which may be deficient in important nutrients like proteins, vitamins and minerals. Beneficiaries may have dietary imbalance as a result. This food is only rich in carbohydrates. The traditional PDS only ensures that all individuals get enough food. They don't consider about the nutrition value. The system might not be able to effectively address certain types of malnutrition. Traditional PDS usually runs independently from other vital domains like healthcare and education. The influence of food distribution on malnutrition may be limited if it is not integrated with health initiatives and nutritional education.

II. LITERATURE SURVEY

The global health suffers from malnutrition globally, especially among vulnerable populations. Kiesswetter et al. stressed the importance of nutrition at older age and impact in healthcare, especially when managing patients with head and neck cancer; it is during this period that dietitians become the first contact for patients suffering from such a condition. So far, the benefits of individualized nutritional counseling would thus outweigh all these barriers, be it patient compliance, resource limitation, or other issues. Similarly, Cook et al. [2] have also emphasized the nutritional challenges a patient with head and neck cancer experiences because of anatomical and functional changes caused by treatment. They emphasize that each patient requires patient-specific support. However, variability in response and limited access to timely dietetic care make the management of such patients challenging. Other than that, a direct relationship between food systems and the environment with malnutrition matters as Fanzo et al. [3] think of calling for interlinked collaboration to foster sustainable food systems in line with the goal of addressing malnutrition while accounting for impacts on the environment.

Balancing, between agriculture and nutrition in terms of sustainability is quite challenging. Regarding socio-economic factors, Singh et al. [4] take up an analysis of child malnutrition in 640 districts of India and report the underweight and stunting prevalence of children in those districts to be more prevalent in the economically less favored districts and largely dependent on the education of their mothers and access to safe water. Panda et al. [5] discuss the linkage between malnutrition, poverty, and efficiency of India's Public Distribution System (PDS), which, though capable of bringing relief in malnutrition, cannot guarantee delivery across all heads and expects vulnerable people to undergo hardships related to malnutrition. From a more macro-level perspective, malnutrition outcomes are also affected by social factors such as governance. According to Biadgilign et al. [6], quality of governance and public health spending are the critical factors mitigating child undernutrition in Ethiopia, though limited data presents a challenge to its clear understanding to be certain about effective governance.

Poverty and malnutrition go hand in hand in a cause-and-effect kind of way, and according to Siddiqui et al. [7], issues need to be dealt with through integrated approaches to break the kinds of cycles of poor health and economic disadvantages.

Climate change adds a further layer of complexity to the fight against malnutrition because, according to Smith and Patel [8], extreme weather conditions, as well as water shortages on top of everything accelerate food insecurity, though it would be challenging to isolate climate's direct effect. ECD programs show promise in improving both physical and cognitive outcomes for children, as argued by Rao et al. [9], though the challenge is seen in implementing the same programs in resource-poor settings. Social safety nets greatly play a role in reducing malnutrition, and cash transfer and food subsidy have been proved to be relatively effective programs, although not sustainable for the long term and potentially leading to dependency, as pointed out by Gupta et al. [10]. Another important relationship that is worth paying attention to is that between nutrition and mental health, which has been described by Williams et al. [11], who illustrated food insecurity and its subsequent impact on mental health outcomes in low-income populations. Although the relationship is well recognized, intervention efforts that are nutrition and mental health multisectoral are relatively quiet narrow in scope.

In that regard, Chawla et al. [12] demonstrate the significance of dietary patterns in relation to the prevention of chronic conditions in populations such as diabetes and hypertension, whereby it can be asserted that healthy diets play a leading role in the management of such risks in vulnerable populations. However, their evidence infers persistent weaknesses in the access of healthy food, particularly in the urban slums. Patel et al. [13] validate the genetic susceptibility of malnutrition-related diseases. The research is on the genetic variations in the metabolization of micronutrients, which may be applied in targeted interventions on a high-risk population. Green et al. [14] discuss the global burden of malnutrition set against the background of pandemics such as COVID-19 and draw implications about how this situation has further strained the prevalence of malnutrition through disorganized food supply chains and healthcare systems. Another important strategy in combating childhood malnutrition is nutrition interventions in schools through inclusion of nutrition education and school feeding programs into the existing structures of schools, as presented by Shah et al [15]. These programs have been shown to be quite effective in improvement in nutritional status but, certainly are not free from implementation challenges such as poor funding and other logistical problems. Another area is in healthcare, as discussed by McGregor et al. [16], which taps the potential of supplementation through routine pediatric care to avoid stunting amongst children under five years. There is further research on micronutrient supplementation programs, by Jadhav et al. [17] who evaluate the effect of vitamin A and

zinc supplementation in reducing child mortality. Their findings focus the problem on the need for micronutrients in combating malnutrition, but the challenge remains in coverage in rural areas. The other area of focus is on dietary diversity and its outcome concerning nutritional health as studied by Kaur et al. [18], who postulate that a greater diversity in diet increases nutrition status in low-income countries. The role of urbanization in dietary habits was analysed by Nguyen et al. while focusing on the transformation toward a diet of more processed food products and implications for malnutrition in highly urbanizing populations.

On the other hand, problems with undernutrition and food insecurity are still prevailing in rural populations and were indicated by Lopez et al. [20]. Zhang et al. [21] highlights the importance of women's empowerment and education on their role in the fight against malnutrition, particularly in rural communities. National Level Policymaking in combatting malnutrition: Murthy et al. [22] examines the national nutrition programs specifically Integrated Child Development Services, India. The work establishes that though ICDS has been effective in combating malnutrition, there remains a lot of scope for improvement regarding implementation. Three papers go towards discussing the role the private sector must play in eliminating malnutrition: those by Jain et al. [23], who study how public private partnerships can facilitate the realization of benefits in food security and nutrition outcomes within the perspective of emerging economies. Gupta et al [24] also evaluate the efforts of food fortification programs toward reducing micronutrient deficiencies by indicating that success in iron, iodine, and vitamin A fortification into staple foods has been achieved, but the problem is overcoming their wide adoption, especially in rural areas.

Singh et al. [25] talked about the extent of mobile technology to promote nutrition education and behavior change. Interventions based on mobile technology could potentially influence diet practices and nutritional education. Harris et al. [26] raised concerns concerning childhood obesity, which exhibited increasing trends over time and was related to various non-communicable diseases in adulthood. Other social determinants of health, including income and education, were identified by Lee et al. [27] as major factors in determining childhood obesity. Nutrition and infectious diseases were discussed by Walker et al. [28], who used the nutrition factor in relation to improve immune function and minimization of susceptibility to infections. Discussion Garcia et al. address the concept of food sovereignty, which has been gaining much attention in the fight against malnutrition; however, it focuses on bringing local control over food systems to ensure equitable access to nutritious food. Nair et al. ends the scientific paper with discussing organizations that are involved in the international world fighting malnutrition, like WHO and UNICEF. They mention efforts to fight malnutrition on a global level through initiatives like SUN.

III. EXISTING METHOD

A. Introduction

In India, the Public Distribution System is the largest food security initiative in the world. PDS was an economic and political tool that the Indian government planned and launched as a part of the national food security program allocating food to each citizen and especially to vulnerable and economically weaker sections. The PDS is an initiative to make essential food items such as rice, wheat, and sugar available at subsidized prices via the Fair Price Shops which are strategically located across the country.

B. History background

PDS had its origin in 1940 when the government was trying to solve food shortages in the country during World War II. Initially, it was a war-time distribution program, but it grew as an all-India program after gaining independence. The PDS was a central component in India's food policy at that time due to its ability to stabilize prices and guarantee the supply of basic commodities during the 1960s and 1970s.

C. Structure and operation of the PDS

The Public Distribution System (PDS) is the one which works with several players in a well-relieved and structured network. The central government is the one that transports, stores, and gives out food grains. FCI is the one that primarily helps in the process of food grain transfers from the farmers to the government under the Minimum Support Price (MSP) scheme. At the state level, the core mandates include the state governments' responsibilities to select people who are entitled, provide food grains to the FPS, and overall PDS administration. FPS are the retail sale depots scattered all over the country which are basically the points of sale from which the beneficiaries can buy food grains at reduced rates. The distribution is done by these through the more than 500,000 FPS covering the whole country. The beneficiaries are also classified based on the economic sections under the TPDS scheme. The two big groups are: the poorest of the poor, the AAY scheme household beneficiaries, who are hence the ones getting 35 kg per month at highly subsidized rates; and Priority Households (PHH), who are benefited with 5 kg of food grains per person per month. This is a very well-organized method to ensure food security to the society's most susceptible members.

D. Limitations of public distribution system in cladding malnutrition in India

The Public Distribution System (PDS) of India functions through a systematic network of stakeholders. The procurement, storage, and allocation of food grains is under the central government's administration, with FCI as its nerve center to acquire food grains at MSP from the farmers. The major task of identifying the beneficiaries, as

well as providing food grains to FPS, rests with the state government. It also monitors PDS at the state level. FPS are the retail points across India that act as sales points where the beneficiary acquires food grains at a subsidized rate. The over 500,000 FPS in India ensure that the distribution reaches all parts of India. The target public distribution system beneficiary is divided into the two lines, TPDS according to the economic status: two main beneficiary categories are the households categorized under the AAY being the poorest of the poor, and entitlements of 35 kilograms monthly at heavily subsidized rates, PHH receiving 5 kilograms monthly for each member. This planning strategy focuses on food security for the weakest members of society.

IV. REFORMED PUBLIC DISTRIBUTION SYSTEM

A. Citizens

The Citizen Module of PDS empowers citizens by providing them with their nutritional status and ration entitlement in real-time. Once a citizen logs into the system, the system shall display information including name, age, family details and registered address of the citizen within the system. This, in turn, allows beneficiaries access to a record of eligibility and displays records of changes done on their profiles. This module allowed citizens to view updated reports concerning health and nutrition. These reports are always updated by the diagnostic centers, and reports might include such important aspects as BMI, iron deficiencies, and vitamin deficiencies, aspects that normally explain the nutritional profile of a citizen. The Citizen Module has again as its prime aim individualized ration subsidy recommendation. Citizens can view from nutritional data which rations they are eligible for. Thus, food distribution is further targeted toward specific nutritional needs. This averts malnutrition mainly and provides for that nutritionally balanced diet, especially in cases where deficiencies are likely to appear in basic nutrients such as proteins, vitamins, and minerals. Such suggestions consider the health condition of every individual so that this distribution system is more responsive and efficient in fighting not only hunger but also hidden hunger. This, at the same time, assures the citizen to receive timely information concerning an important event such as a medical camp, collection of rations and scheduling, and changes in eligibility or ration allotment. In this context, this feature updates the citizens about the medical camps in which they are permitted to undergo health checkup besides stating when and where they can take their rations. This makes citizens more attentive towards participation in the PDS system and increases overall accessibility as well as user-centric use of the system. Through this excellent integration of health and nutrition data, the Citizen Module takes on the prominent role of augmenting food security and, consequently health benefits, for the disadvantaged population groups.

B. Central administration

This is the central controlling unit for the whole Public Distribution System (PDS). It works towards an effective control over citizen data, ration distribution, and general working of the system. Central Admin possesses the powers of handling the citizen and ration shop, add, modify, or delete data if needful. It will verify current profiles like a change in family size or address by accepting/rejecting applications based on proper verification. Data is current and accurate. It ensures the distribution of rations to shops, keeping in mind the stock levels and citizen's nutritional needs through fair and efficient allocation of resources. One of the other functions of the Central Admin Module is that it decides for medical camps to be undertaken periodically for citizens; normally it is done once every three months. These camps have proved to be a very important method of monitoring and administering remedy measures against malnutrition by making health checks and updating the nutritional records of the citizens. The Central Admin informs all the diagnostic centres and citizens of the camps they can participate in so that as many people as possible participate. This module further helps to track nutritional deficiencies and health trends in making the best decision of rations to be administered by the admin team. The report from these health records will highlight where malnutrition is rampant, so the resources must be directed towards the more vulnerable populations. Central Admin receives a full range of analytical tools and statistics of reports regarding trends of malnutrition, efficiency in ration distribution, and nutritional deficiencies within different regions. Using such analytics, decisions are taken in data-driven scenarios to optimize the system, enhance food distribution strategies, etc., all through solutions that directly impact public health. The module also enables the central team to monitor the effectiveness of the rationing process and adjust where necessary to ensure that the PDS operates at its maximum efficiency. Generally, the Central Admin Module provides the center with a very vital role in managing and controlling the operational side of the PDS, with emphasis on food distribution equitably and addressing malnutrition at national levels.

C. Ration Shop

The module on Ration Shops is one of the most important modules of PDS, ensuring the proper dissemination and distribution of food grains and other significant commodities to eligible citizens. Ration shops facilitate the mechanism of disbursement after receiving an allocated number of rations through the central system. Each ration shop holds a list of registered citizens and the rations they must receive based upon their nutritional needs. Such data can be verified by the Ration Shop In-Charge, eligibility criterion cross verified, and rations accordingly distributed. It also helps in updating the status of ration distribution at a ration shop. For example, has a citizen or not collected his allocated rations or not? Distribution. The ration shop plays an important role in ensuring that such distribution should

not have any harassments and should be transparent in distribution.

They can thus prepare distribution reports reflecting the volume of ration that goes out in a particular period. These reports are highly important in auditing and while tracking the overall performance of the PDS. Ration shops may send notices to citizens upon availability of rations, a change in schedule for ration distribution, or even reminders to collect their rations. It keeps the citizen informed so that he does not miss out on essential food supplies. Ration shops also become responsible for authenticating eligibility of citizens towards receiving rations. This includes the mechanisms to verify numbers such as income levels, family size, and so on defined by the central administrator. This verification ensures that only those eligible persons get their subsidized food grains, thereby curbing the extent of malpractices or fraud. The Ration Shops Module provides the management authority applications for this process to be operated effectively to help it being successful for the whole PDS.

D. Diagnostic centres

This module of the Diagnostic Centre helps monitor and track the nutritional health of the people through the PDS. The diagnostic centers will be expected to undertake check-ups on the health of people. This way, the history of the nutritional status of a particular person shall be made. Upon completing the above check-up of health including assessing Body Mass Index, iron checking, vitamin deficiencies, and others, the diagnostic centre input such information in the system. This also helps the central system update records on citizens' health and determine specific nutritional requirements. The system utilizes processed data from uploads to produce reports on nutritional statuses and wisdom on the citizen's health. Such reports allow for the decision on the ration distribution to ensure that citizens get just the right quantities of food based on health conditions. For instance, acute nutritional deficiency patients can be provided with surplus foodgrains or individual items that can be part of medical treatment.

The health records of these patients would also periodically be updated under the module of Diagnostic Centres, especially after health camps where health data comes in real-time and thereafter can be used in effective ways to sound rationing decisions. In addition to health monitoring, the diagnosing centers are also doing a wonderful job regarding citizen knowledge about their health condition. After uploading and processing data, citizens can view record updates on health and watch their changes over time. This helps citizens have an enabling approach towards health improvement and nutrition. It allows the general trends within populations so that the central admin and health authorities can focus on those with increased rates of malnutrition. The Diagnostic Centres Module thus becomes important in keeping health information updated, resource marshalling being proper, and the efficiency of the PDS in checking malnutrition.

E. Recommendation System

1. Basic Food Allocation: The Basic Food Allocation refers to a free food distribution system targeting the allocation of basic nutrients to every citizen, irrespective of their income level, in a bid to ensure that everyone has what is needed to maintain healthy living. These foods are free and set according to one's age and income levels.

1.1 Low-Income Group: For the Low-Income Group, the minimum food grant focuses on providing nutrient-rich, affordable foods that help fulfill essential nutritional requirements like vitamins, minerals, and macronutrients. This group receives foods that provide crucial calories and nutrients necessary for growth and development.

- *Children (0-7 years):* Foods such as quinoa, cornmeal, oats, and iodized salt are provided to small children. These are essential for early childhood development, helping them grow and develop, covering energy, vitamins, and minerals like iodine and iron.
- *Adolescents (8-20 years):* Teenagers in this group receive slightly higher amounts of foods like quinoa, cornmeal, oats, and iodized salt. During this fast growth phase, they require more iron, zinc, and vitamins to support energy, immunity, and physical growth.
- *Adults and Elderly (21-100 years):* The staple share for adults comprises quinoa, cornmeal, oats, and iodized salt. For the elderly, these foods provide energy while addressing micronutrient deficiencies, especially iron and calcium, which are more prevalent among the aged. The aim is to ensure proper growth in childhood, development in adolescence, and health throughout adulthood and old age.

1.2 Middle-Income Group: The Middle-Income Group receives a balanced supply of foodstuffs, providing the ability to afford specific types of food that fill nutritional gaps. While middle-income families still have some buying power, they benefit from subsidies that support more expensive foods.

- *Children (0-7 years):* This group receives basic foods like brown rice, wheat, oats, and iodized salt. These foods contain sufficient calories and vitamins for growth.
- *Adolescents (8-20 years):* Teenagers and young adults in this category receive generous portions of brown rice, wheat, oats, and iodized salt. These foods provide the necessary carbohydrates and proteins to sustain their growth and activity.
- *Adults and Seniors (21-100 years):* For adults and seniors, their staple food allocation includes brown rice, wheat, oats, and iodized salt, ensuring basic energy needs are met and helping maintain health, reducing risks of chronic conditions like hypertension and diabetes. The diversity of food

helps support balanced nutrition for all age groups.

1.3 High-Income Group: For the High-Income Group, the system ensures that they also receive enough basic staples, though the quantity provided is typically less than for low- and middle-income groups. This group focuses on wholesome, well-balanced diets using affordable staple foods.

- *Children (0-7 years):* High-income children receive foods like rice, wheat, oats, and iodized salt. These foods meet the caloric and nutritional needs of young children at this income level.
- *Adolescents (8-20 years):* Teenagers in this group are provided with rice, wheat, oats, and iodized salt, ensuring they get all the essential nutrients, though in smaller proportions compared to the lower-income groups.
- *Adults and Seniors (21-100 years):* High-income adults and seniors receive rice, wheat, oats, and iodized salt as their basic staples. Since they have better disposable incomes, these foods help keep them healthy without relying heavily on subsidies, ensuring the efficient use of public resources.

2. Provision at Cost: In addition to the free Basic Food Allocation, the Paid Food Allocation caters to nutrition-rich, high-value foods that citizens can buy at subsidized rates. These foods help fill specific nutritional gaps and support a balanced diet, with prices determined by nutrient density and availability.

2.1 Low-Income Group: For low-income citizens, the paid food allocation enables access to higher-quality foods at subsidized prices. These foods include sources of protein, healthy fats, and micronutrients, which might otherwise be unaffordable. Examples include potato, sunflower oil, chickpeas, and peanuts, all of which are rich in fiber, healthy fats, and proteins that help reduce malnutrition and promote overall health.

2.2 Middle-Income Group: For the Middle-Income Group, the subsidized foods are relatively expensive but available at reduced prices. These foods include wheat bread, sweet potatoes, soybean, and milk powder, ensuring families get essential proteins, vitamins, and micronutrients to support a balanced diet and replenishment of their nutritional needs.

2.3 High-Income Group: For the High-Income Group, the paid allocation includes premium, nutrient-dense foods that are higher in cost. These include cashews, dry grapes, pumpkin seeds, and sunflower seeds. These foods are packed with healthy fats, vitamins, and antioxidants, designed to ensure well-being and disease prevention among adults who can afford such higher-end nutritional items.

3. Combining Basic and Paid Food Allocations: The combining of paid food allocations with basic food

allocations involves various approaches for nutrition, efficiency, and equity in food distribution. Through the combination of basic and paid food allocations, the system ensures that all people have nutritionally adequate diets with fairness across all levels of income.

3.1 Credit-Based System: In the process of the delivery system for paid food allocations, a credit-based system is a core part of the system. To each citizen he would give credits according to his income level and health needs. A lower income citizen would be assigned more credits, thus ensuring that those with a lower income are better placed to afford nutrient-dense food at subsidized rates. This, therefore, means the rich will be less likely to avail of the subsidy assigned to them since they are assigned lesser credits. It helps prevent the misallocation of public funds and serves to focus assistance on those in the highest need. Low-income people receive the greatest number of credits to purchase nutrient-dense foods more cheaply. Middle-income consumers receive a moderate number of credits, which they use to buy a better balance of staple and healthier foods. High-income people receive fewer credits because they can purchase most of their foods without assistance. Vital Emergency Health Needs The system also selects food items based on nutritional deficiencies to meet the needs concerning current health issues through health data. Food items meant for citizens suffering from specific health issues such as anemia, vitamin deficiencies, and malnutrition are identified and issued to these citizens through relevant food items. Quarterly health data collected by diagnostic centers are used to ensure the system recommends foods appropriate for a citizen's present nutritional needs. For example, when one has low levels of iron, soybeans or chickpeas would be recommended, as they are full of iron. Real time coverage in deficiency is ensured by the system through recommending foods that can improve the health of the individual step by step.

3.2 Income-based Priority Food: Accounting for income distribution, the first characteristic of this recommendation system has high-priced food containing either high calories or nutrients to be given to the rich, while the poor will have cheaper food that is rich in nutrients, which fulfills basic needs towards health status. Nutrient-rich, high-value foods will be focused on for low-income groups, which are fundamental to fill nutritional gaps and enhance the status of health. Medium-income consumers are offered a choice between two staple food products that cost less but contain the required nutrients, thus ensuring them nutritional content at reasonable cost control. High-income consumers are subsidized for fewer staple food items; they can choose to buy the very expensive cashew nuts, dry grapes, or sunflower seeds at a cheaper price just due to its healthy fats and nutrient content.

3.3 Reductions in the Number of Food Subsidies: The system limits food items to a total of seven per citizen to simplify the distribution process as well as to avoid distributing excessive food items, ensuring that food distributions remain manageable and that the nutritional balance is put in place. There is the basic allotment that has

staple products like rice, wheat, cornmeal, and iodized salt amongst other essential products. After the basic provisions, the available spaces are then filled with foodstuffs in coordination with the citizen's income and level of health. Such allocations see that every individual gets wholesome food covering short-term and long-term nutritional needs.

3.4 Adjustments Every Quarter: The system of food rationing is continually revised quarter by quarter to define changes in health conditions and any adjustment in the income level in making food distribution. The country reviews and adjusts the strategy of allocation for food every quarter due to the refreshment of health metrics plus any modification of a citizen's source of living. Records of health information gathered from diagnostic centers every three months guide the allocation process in identifying any emerging deficiency or health condition that needs attention. Changes in income levels or family circumstances are also considered, so that the allocation remains fair and relevant to the current situation of each citizen.

V. IMPLEMENTATION

Chapter 1: Analysis and design of the system

1.1 Problem Statement

The Public Distribution System (PDS) is a government initiative designed to provide subsidized food grains to low-income households in India. Despite its success in reducing hunger, PDS has not fully addressed the issue of malnutrition, particularly the deficiencies in essential nutrients faced by many individuals. The aim of this project is to integrate the existing PDS infrastructure with personalized nutritional recommendations and health data, improving food distribution based on nutritional needs and thus addressing both hunger and malnutrition.

1.2 System Requirements and Scope

The proposed system consists of several modules: the Citizen Module, which allows citizens to access their personal and nutritional information, receive ration recommendations, and track health data; the Central Admin Module, which provides administrators with tools to manage citizen data, ration allocations, and medical camp organization; the Diagnostic Centres Module, which enables healthcare facilities to upload citizens' health reports, including nutritional deficiencies; and the Ration Shops Module, which allows ration shops to verify eligibility and distribute food. The system will provide functionalities such as real-time health updates, ration distribution schedules, and notifications about medical camps. The goal is to ensure citizens receive food tailored to their nutritional needs.

The system follows a modular architecture with a frontend, backend, and database layer. The Frontend Layer includes web interfaces built using React.js, providing an interactive and dynamic user experience. The Backend Layer is developed with Python, utilizing frameworks like Flask for handling API requests, business logic, and communication with the database.

The Database Layer uses PostgreSQL, storing citizen data, health records, ration distribution information, and reports. The system ensures secure authentication and authorization, allowing different roles (admin, citizen, ration shop) to access appropriate functionalities.

1.4 Functional Design

The Citizen Module allows citizens to log in, view their health status, track nutritional deficiencies, and receive ration recommendations based on health data. Citizens can also update personal information, view ration schedules, and attend medical camps. The Central Admin Module provides administrators with the ability to manage citizen data, approve ration allocations, and organize health-related initiatives.

Admins can generate detailed reports on malnutrition trends and track the system's effectiveness. The Diagnostic Centres Module allows health facilities to upload diagnostic reports, including nutritional deficiencies, which will influence ration allocations. The Ration Shops Module helps ration shops verify eligibility and manage food distribution.

1.5 Database and Technology

The system utilizes a robust database design and modern technologies to ensure efficient operation and scalability. The database schema includes several key tables: Citizen Data, which holds personal details, health data, ration eligibility, and allocation history; Ration Data, which tracks the types, quantities, and distribution schedules of food rations; Health Records, which stores diagnostic data such as nutritional deficiencies and medical history, updated by diagnostic centres; and Reports and Analytics, which stores generated reports related to malnutrition trends, ration distribution efficiency, and citizen feedback.

This database is designed for data integrity, fast retrieval, and seamless interactions between modules. For the frontend, React.js is used to build a dynamic and responsive user interface, while the backend is powered by Python with frameworks like Flask, enabling the development of scalable REST APIs and handling business logic. The system's PostgreSQL database ensures reliable and efficient data storage and management, while a notification service alerts citizens about their ration schedules, eligibility changes, and health updates.

2.1 Architecture Design

The system follows a modular architecture, dividing the functionality into four main modules: Citizen Module, Central Admin Module, Diagnostic Centres Module, and Ration Shops Module. This design enables scalability and maintainability. The Frontend Layer is responsible for rendering dynamic content and handling user interactions. React.js is used to ensure smooth, interactive communication between the user and the system. The Backend Layer is powered by Python and connects to the frontend and database, processing user requests and performing necessary computations. The Database Layer uses PostgreSQL to store all necessary data, ensuring consistency and fast data retrieval.

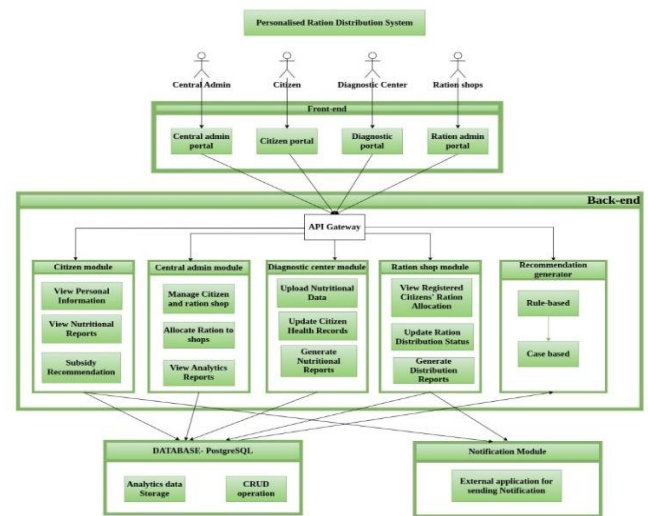


Figure 1: Architectural Diagram

2.2 Functional Design

The Citizen Module allows citizens to register, log in, and access their health and ration data. It displays personalized nutritional recommendations based on their health data and eligibility. The Central Admin Module enables admins to manage citizens' data, approve ration allocations, and generate reports on nutrition trends. The Diagnostic Centres Module allows health centres to upload data about nutritional deficiencies and health conditions. This data helps generate personalized ration recommendations. The Ration Shops Module assists ration shops in verifying eligibility and distributing rations efficiently to eligible citizens.

2.3 Database design

The database schema is designed with several key tables to ensure efficient data management and interaction across modules. The Citizen Table stores essential information such as name, age, household size, health status, and ration allocation history. The Ration Allocation Table tracks the

types, quantities, and distribution dates of food rations provided to citizens. The Health Records Table stores health-related data, including nutritional deficiencies and chronic conditions. Lastly, the Report Table contains reports on malnutrition trends, ration distribution, and citizen feedback. These tables are interrelated to facilitate smooth data retrieval, ensuring seamless communication between modules and efficient operation of the system.

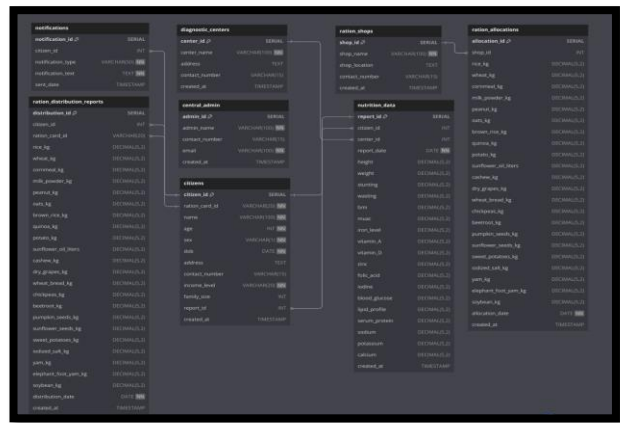


Figure 2: Database design Diagram

Chapter 3: Coding and implementation

3.1 Core Implementation

The core implementation of the system focuses on the seamless integration of the frontend, backend, and database to provide a smooth user experience and efficient functionality. For the frontend, React.js is used to create a responsive and dynamic user interface. React components handle different roles in the system, such as citizen information display, ration allocation management, and health report tracking. Backend development utilizes Python with Flask (or Django) to create RESTful APIs, which interact with the frontend and manage business logic. The backend handles citizen data processing, health record updates, ration distribution logic, and notifications. The database, PostgreSQL, stores all critical data, including citizens' personal and health information, ration distribution details, and analytical reports.

3.2 Implementation Screenshots:

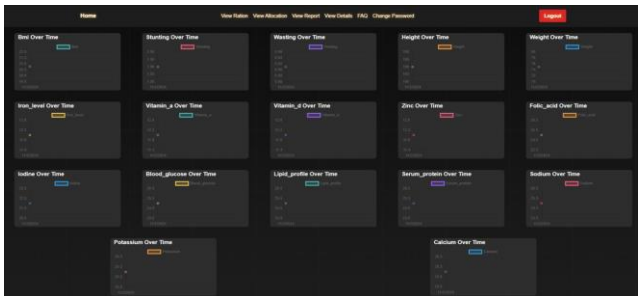


Figure 3: Citizen Dashboard

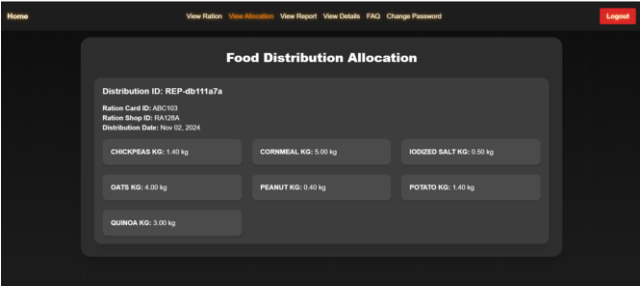


Figure 4: View Allocation

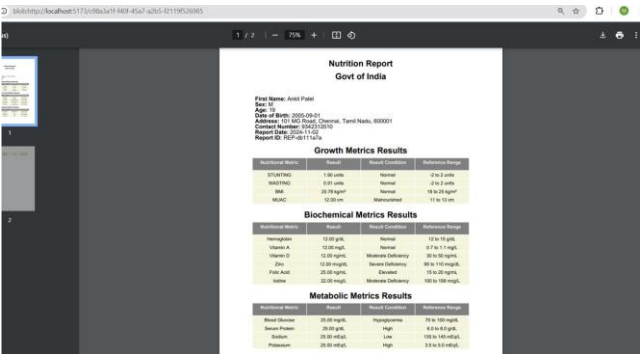


Figure 5: View report in PDF

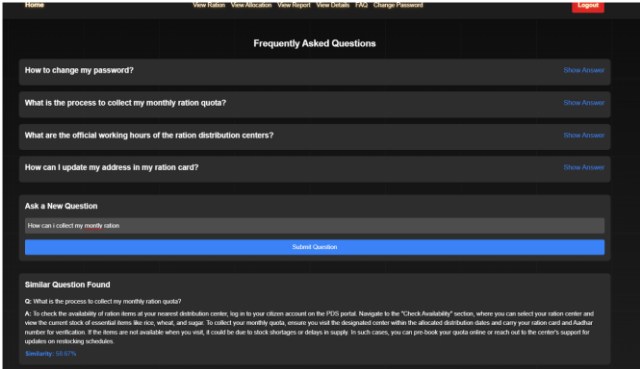


Figure 6: FAQ with case-based system

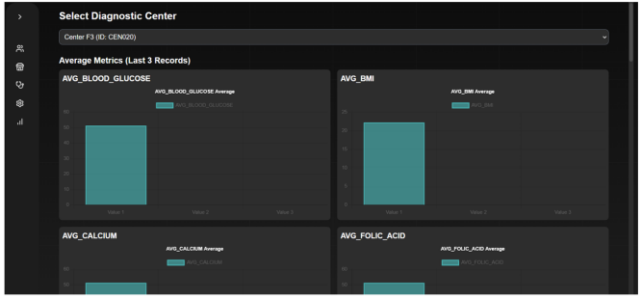


Figure 7: Comparative analysis of metrics

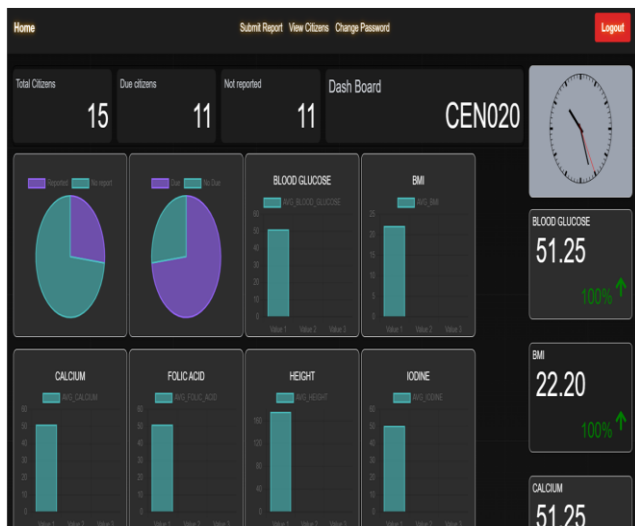


Figure 8: Dashboard of Diagnostics centre

Chapter 4: Fitting into society

4.1 Understanding your societies' needs and context

In a Personalized PDS, the needs and context of your society will be understood. Do a detailed demographic analysis of the population to understand size, age, and the level of income, as well as how many vulnerable groups, such as children, pregnant women, and the elderly would be. Conduct a health and nutrition survey to assess common deficiency and malnutrition rates in areas. This will help the fine-tuning changes about the system and health/food security concerns. Review and assess the ration distribution system as it is, its strengths, and areas for improvement and how this could make the changeover to the new system easier and smoother.

4.2 Identifying Stakeholders and Involving Them

It is equally as important that the approach involving stakeholders in the implementation is most effective. Cooperating with local government officials who are responsible for accessing allocations, funds, and policy support. Should involve all community leaders and NGOs to create trust and deepen the system among citizens and thereby influence their community. Should deploy health service providers and diagnostic centres to undertake health assessments and update nutritional data. Ration shop owners must be trained effectively for the new system. Lastly, engage the citizens at the front-end very early on through workshops and consultation to satisfy the needs and grievances that would inform ownership by the community.

4.3 Technology Infrastructure

There is a robust technology back end to support Personalized PDS. The Citizen Module will be an interface through mobile application and web portal, to view health

reports, ration entitlements, and filing grievances. The real-time health data upload module will be implemented so the nutritional profile would be available in real-time. The citizen biometric-based or QR code-based verification along with real-time inventory management would be really important for the Ration Shop Module. The Central Administration Module would centralize data helping policymakers spot trends and make informed decisions based on it. Thus, this well-integrated system would further ensure smooth working and transparency as well.

4.4 Pilot Test of the System

Before implementing it on an all-India scale, a pilot test must be conducted in specially chosen areas. A pilot test of the system in a number of localities-known to be urban, rural, and semi-urban-would show its adaptability in different places and effectiveness under varied situations. All users, citizen and ration shop owner and health worker should be exposed in the pilot phase to the new system. Monitor the pilot very closely and observe technical and operational issues that point out but also solicit the feedback of all stakeholders. Refine the system on these bases not to manifest extreme problems in rollout.

4.5 Scaling and Scaling Up

Roll out when proven effective. Roll out the system gradually to the entire society, beginning from stage one, learning and improving at each and every step. The system has to be accessible to all people while offering multi-lingual support and work offline for remote locations. Have a helpline network with teams on the ground to facilitate people in their transition into the system. This will minimize disruption to society and ensure an orderly roll-out process.

4.6 Continuous Improvement

Since Personalized PDS is a dynamic system, it cannot be static. Firm feedback mechanisms should be established wherein citizens and other stakeholders may raise their concerns and propose improvements to the system. Advance types of data analytics have to be developed for tracking trends and measuring the impact of the system and identifying fields to be improved. It will update regularly based on changing scenario with technological advancement and social needs. The programs of continuous training and capacity building for all users will also help in sustaining optimum levels of efficacy as well as satisfaction levels.

4.7 Public Awareness and Community Involvement

It is the public awareness that acts as the backbone of such a system. Carry out routine public awareness through social media, local community events, and workshops, making the citizenry aware of the benefits and functionalities of the

system. Organize camps so all citizens participate actively, use the system, and get involved in the decision-making process. Invest in NGOs and maintain its outreach at the grassroots level to reach the most vulnerable populations and supplement areas of need. This system will become more effective when it builds a sense of responsibility and participation among the community.

4.8 Governance and Oversight

To bring transparency and accountability under one umbrella, strong governance and oversight are necessary. Designate a monitoring committee of different stakeholder groups to oversee the operations of the system. Conduct frequent audits in ration distribution and health reports to find out fraud and any kind of non-compliance with policies established. Review and update policies through periodic review by the use of performance data, emerging health trends, and feedback from stakeholders. The integrity of the system shall continue to serve the community effectively.

VI. TEST CASES

For example, a single citizen entry named Arjun Kumar, based on his health metrics, the personalized food allocation aims to address his specific nutritional needs. His growth metrics, such as stunting (0.0 units), wasting (1.5 units), BMI (22.50 kg/m²), and MUAC (12.5 cm), all fall within normal ranges, indicating no immediate issues with his growth or nutritional status. However, his zinc levels are slightly deficient, and his calcium and sodium levels are below the recommended ranges, requiring special attention to those deficiencies. The food allocation includes rice (5.00 kg) and wheat (5.00 kg), both of which provide essential carbohydrates and energy. These are chosen to maintain Arjun’s overall energy levels, which are necessary for optimal functioning and daily activities.

Rice is a primary source of energy, while wheat contributes to fiber and B vitamins, which are vital for his metabolism and digestive health. Milk powder (1.00 kg) is allocated to help address his low calcium levels. Calcium is critical for bone health, and milk powder also provides vitamin D, which is necessary for calcium absorption. This combination ensures that Arjun can maintain strong bones and support his immune system. Oats (3.00 kg) are also included as a rich source of fiber, especially beta-glucan, which helps regulate blood cholesterol and support heart health. Oats also contain iron, contributing to his overall micronutrient intake. To further address his zinc deficiency, chickpeas (1.00 kg) are allocated. Chickpeas are an excellent source of zinc, which plays a critical role in immune function, protein synthesis, and cell growth. Additionally, iodized salt (0.50 kg) is included to support his iodine levels, which are slightly below the normal range. Iodine is essential for thyroid function, and this small addition of salt ensures he receives the necessary iodine to support his metabolism. This personalized food distribution plan ensures that Arjun Kumar receives a

balanced and nutrient-dense diet tailored to his specific health needs, targeting his deficiencies while maintaining overall nutritional stability.

Metrics	Result	Condition	Reference Range
Stunting	0.0 units	Normal	-2 to 2 units
Wasting	1.5 units	Normal	-2 to 2 units
BMI	22.50 kg/m²	Normal	18 to 25 kg/m²
MUAC	12.5 cm	Normal	11 to 13 cm
Haemoglobin	12.5 g/dL	Normal	12 to 15 g/dL
Vitamin A	1.0 mg/L	Normal	0.7 to 1.1 mg/L
Vitamin D	40 ng/mL	Normal	30 to 50 ng/mL
Zinc	70 mcg/dL	Mild deficiency	90 to 110 mcg/dL
Folic Acid	18 ng/mL	Normal	15 to 20 ng/mL
Iodine	150 mcg/L	Normal	100 to 199 mcg/L
Blood Glucose	85 mg/dL	Normal	70 to 100 mg/dL
Serum Protein	6.5 g/dL	Normal	6.0 to 8.0 g/dL
Sodium	130 mEq/L	Low	135 to 145 mEq/L
Potassium	4.2 mEq/L	Normal	3.5 to 5.0 mEq/L
Calcium	7.0 mg/dL	Low	8.5 to 10.2 mg/dL
Cholesterol	180 mg/dL	Normal	200 to 239 mg/dL

Table 1: Health metrics

VI. RESULTS AND DISCUSSION

The integrated key health metrics into Personalized PDS would be taken, which includes BMI, MUAC, blood glucose, serum protein levels, and various levels of other micronutrients, like vitamin A, zinc, and iodine, with respect to the nutritional needs of each individual. Considering the metrics would help the system provide specific food subsidies addressing both macro and micronutrient deficiencies for each of the individual conditions hence critical for the reduction of improved nutritional status of malnutrition risk. This serves to directly contribute towards Sustainable Development Goal 2, SDG 2: Zero Hunger, especially target 2.2, ending all forms of malnutrition, by 2030. The PDS, however, is more specific than typical mass public distribution systems that primarily rely on general classification criteria such as income or age groups. It incorporates more specific health statistics, and it can, therefore target at-risk individuals

more accurately, not only those undernourished by the standards of body weight or height, but also those who suffer from micronutrient deficiencies or imbalances in metabolic metrics. Thus, expenditure on food intakes and consequently its wastage can be minimized, and health outcomes can also be enhanced. The other strong advantage of this system is the consideration of income with the health data. In this, the low-income populations, which are confronting the most significant nutritional challenges, get more allocations of nutrient-dense food, such as protein and iron-rich foods. The approach ensures that the most nutritionally vulnerable populations receive support that can better enhance their overall health and reduce health disparities in accordance with SDG target 10.2, as it seeks to empower and promote the social, economic, and political inclusion of all. Through the targeting of the most nutritionally vulnerable individuals, it further contributes to reducing inequality and advancing social inclusion. In addition, information about macronutrients and micronutrients forms part of the system to ensure the rationed food given adheres to the health needs of an individual with respect to age, hence another innovative step toward effective solutions in meeting the problem of malnutrition. It also opens the possibility of real-time health tracking, thus making it viable for real-time adjustment in the food subsidies being given based on the evolution in the health status, such as changes in blood glucose levels, potassium, or sodium imbalance.

It follows that dynamic tracking would mean that the entire food distribution at all times is streamlined to reflect what an individual's current nutritional status should be, to optimize health results and help achieve the SDG target 2.1-access of all people, especially the poor and the vulnerable, to safe, nutritious, and sufficient food all year round. Although the system appropriately uses the available health data, integration of longitudinal health data is necessary to achieve further personalization of allocations and refine the approach with the evolution of health changes over time. This would, therefore go hand-in-hand with the dynamic subsidy factor, which would happen at a greater frequency, adjusting itself according to specific health parameters such as vitamin D or iron levels, in order to ensure food delivery is synchronized with the real needs of the individual's health. With fine-tuning of personalization and the real-time adaptation of the food subsidy allocation, the system could maximize impact towards improvement of public health and reduction of malnutrition, thereby further contributing to SDG 2, as well as other related goals on health and wellbeing.

VII. CONCLUSION AND FUTURE ENHANCEMENTS

The PDS proposed is certainly a quantum leap for the alleviation of malnutrition and improvement of public health through the most personalized and data-led approach to food subsidy allocation. It would integrate detailed health metrics-bio metric summary, including BMI, MUAC, blood glucose, serum protein, and micronutrient levels (vitamins, zinc, iodine, etc.) into subsidy determination, so that subsidies depend on the fact that

there is a right person facing deficiency. This development does not only make the process of distributing food more efficient but also enables to achieve SDG 2 – Zero Hunger in terms of both macronutrients as well as micronutrients. The income, along with the health data, is taken into account by the system; therefore, it permits more nutrient-rich food to the most nutritionally vulnerable, in other words, the low-income sections of society.

This would also minimize health inequalities as well as ensure equal distribution of subsidies - SDG 10 (Reduced Inequality). Additionally, with this strategy for allocation based on the existing status of health of an individual, this system opens itself up for real-time monitoring of health and changes in health conditions form a basis for changes in food allocation to optimize it further as well. This system still has scope for further improvement. Aggregating health data longitudinal may only culminate into higher personalization to monitor the health over time and improve food subsidy suggestions even further. This method updates dynamically the subsidy based on health metrics such as vitamin D or iron levels will be an avenue to enhance accuracy and the timeliness that is implemented in the disbursement.

This would thus make it ensure uniformity in the satisfaction of nutritional needs, along with changes in health status. For future development, the system can be so designed to predict health needs in the future using the trends and history data, which will offer proactive intervention before the deficiency becomes critical. The system can further be extended to regional diet variations and health conditions to help better the overall health of diverse populations especially in rural and underserved regions. Conclusion In summary, the Personalized PDS is a more revolutionary concept of public health nutrition, with targeted food distributions by cluster of interest and health status as well as income status. With more dynamic health data to fuse in and interactive predictive capabilities, the overall potential for the system may be nothing less than a drastic decrease in malnutrition, improved health outcomes, and, in all senses, contributions to the realization of SDGs.

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